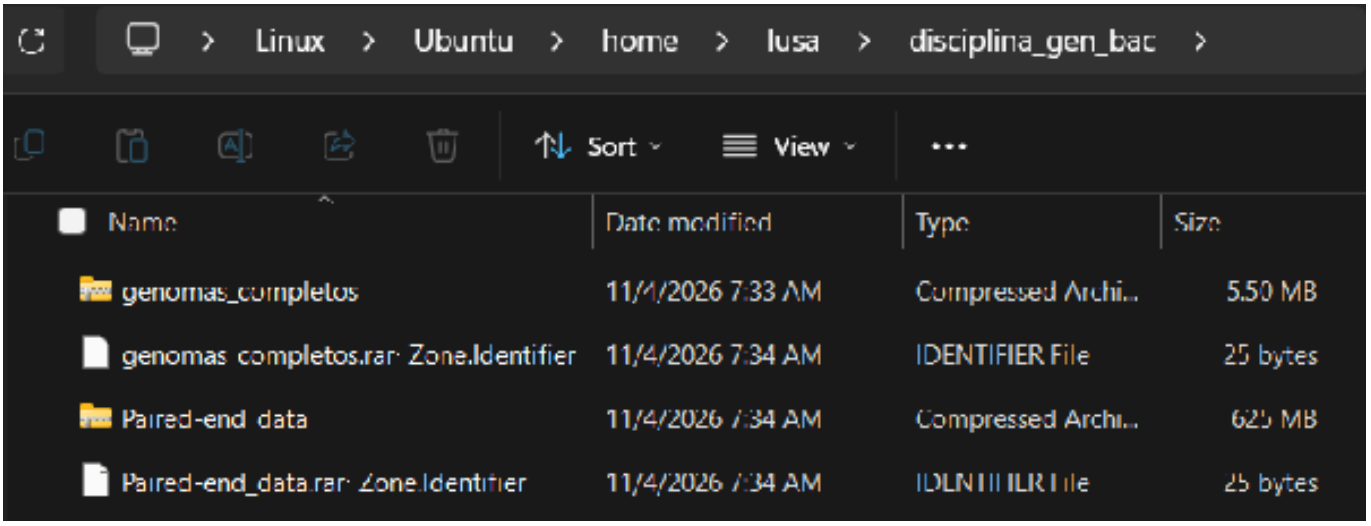
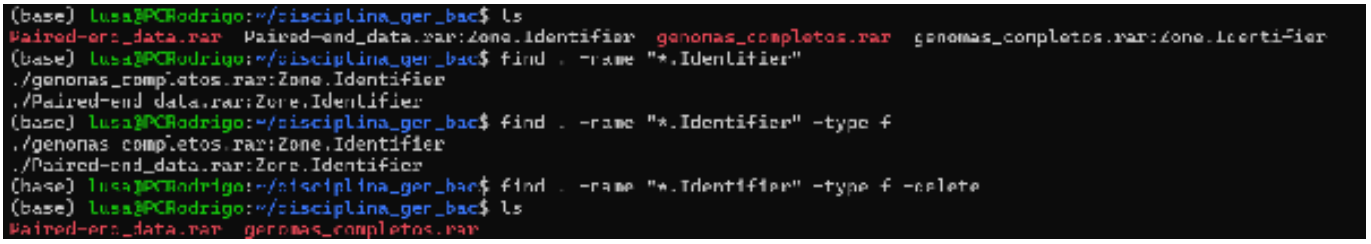
# MONTAGEM DE GENOMAS

## Preparando os Datasets do caso clínico

1. Entre no Google Drive [clicando aqui]
2. Baixe o arquivo `paired_end_data.rar`
3. Baixe o arquivo `genomas_completos.rar`
4. Cole os arquivos em `home/devuser/disciplina_gen_bac`




5. Depois rode os comandos um por linha dando enter:

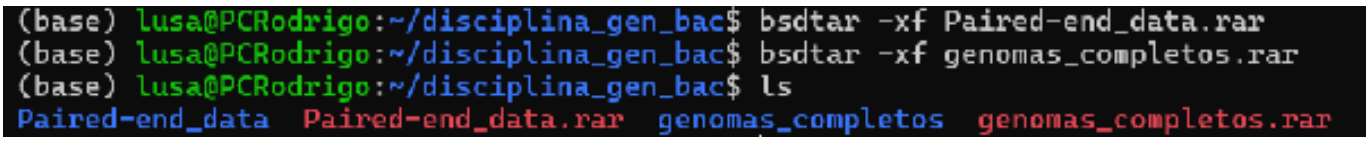



6. Extraia os arquivos usando bdstar:

`bsdtar` = Programa utilizado para manipular arquivos compactados (tar, rar, zip etc.).

`-x` = Extract: extrair os arquivos.

`-f` = File: indica que o próximo argumento é o nome do arquivo.




7. Remova os arquivos zipados

```
(base) lusa@PCRodrigo:~/disciplina_gen_bac$ ls Paired-end_data
pc1  pc2  pc3  pc4
(base) lusa@PCRodrigo:~/disciplina_gen_bac$ ls Paired-end_data/pc1/
forward.fastq.gz  reverse.fastq.gz
(base) lusa@PCRodrigo:~/disciplina_gen_bac$ ls Paired-end_data/pc2/
forward.fastq.gz  reverse.fastq.gz
(base) lusa@PCRodrigo:~/disciplina_gen_bac$ ls Paired-end_data/pc3/
forward.fastq.gz  reverse.fastq.gz
(base) lusa@PCRodrigo:~/disciplina_gen_bac$ ls Paired-end_data/pc4
forward.fastq.gz  reverse.fastq.gz
(base) lusa@PCRodrigo:~/disciplina_gen_bac$ ls genomas_completos
pc1.fa  pc2.fa  pc3.fa  pc4.fa
(base) lusa@PCRodrigo:~/disciplina_gen_bac$ rm -rf *.rar
(base) lusa@PCRodrigo:~/disciplina_gen_bac$ ls
Paired-end_data  genomas_completos
```

O diretório deve estar assim

```
(base) lusa@PCRodrigo:~/disciplina_gen_bac$ tree
.
├── Paired-end_data
│   ├── pc1
│   │   ├── forward.fastq.gz
│   │   └── reverse.fastq.gz
│   ├── pc2
│   │   ├── forward.fastq.gz
│   │   └── reverse.fastq.gz
│   ├── pc3
│   │   ├── forward.fastq.gz
│   │   └── reverse.fastq.gz
│   └── pc4
│       ├── forward.fastq.gz
│       └── reverse.fastq.gz
└── genomas_completos
    ├── pc1.fa
    ├── pc2.fa
    ├── pc3.fa
    └── pc4.fa

7 directories, 12 files
```

## Controle de qualidade das *Reads*

https://github.com/OpenGene/fastp

1. Instalar (uma vez) - ignore se já foi instalado:

```
mamba create -y -n fastp -c conda-forge -c bioconda fastp
```

2. Ative o ambiente conda

```
(base) lusa@PCRodrigo:~/disciplina_gen_bac$ conda activate fastp
(fastp) lusa@PCRodrigo:~/disciplina_gen_bac$ fastp
fastp: an ultra-fast all-in-one FASTQ preprocessor
version 1.3.2
usage: fastp [options] ...
options:
  -i, --in1                            read1 input file name (string [=])
  -o, --out1                           read1 output file name (string [=])
  -I, --in2                            read2 input file name (string [=])
  -O, --out2                           read2 output file name (string [=])
```

3. Criando os diretórios

```
mkdir -p fastp_input fastp_output resultados/01_fastp scripts
```

4. Baixando o script do fastp para rodar em paralelo

```
wget -O scripts/parallel.py \
https://raw.githubusercontent.com/OpenGene/fastp/master/parallel.py
```

5. Reorganizando as pastas

```
for sample in pc1 pc2 pc3 pc4
do
    ln -s "$(pwd)/Paired-end_data/${sample}/forward.fastq.gz" \
          "fastp_input/${sample}_R1.fastq.gz"

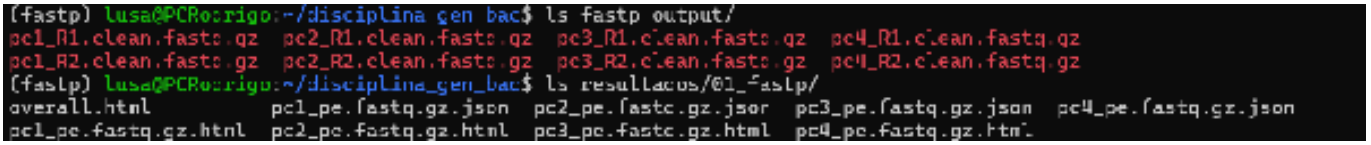
    ln -s "$(pwd)/Paired-end_data/${sample}/reverse.fastq.gz" \
          "fastp_input/${sample}_R2.fastq.gz"
done
```




6. Rodar o fastp em paralelo:

```
conda activate fastp
python scripts/parallel.py \
-i fastp_input/ -o fastp_output/ -r resultados/01_fastp/ -1 R1 -2 R2 -p 4 \
-a "--thread 1 --detect_adapter_for_pe --trim_poly_g --cut_front --
```

```
cut_right --cut_right_mean_quality 20 --qualified_quality_phred 20 --
unqualified_percent_limit 30 --length_required 50"
```




### Perguntas

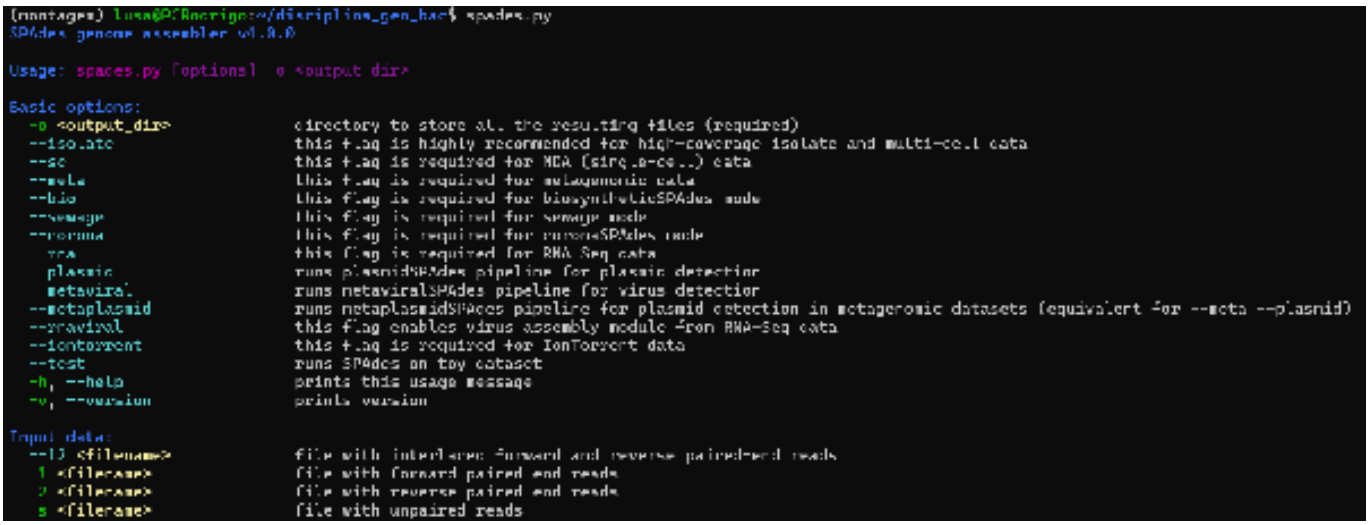
1. As reads têm qualidade boa o bastante para montar os genomas?
2. Ouve um aumento de qualidade pós QC?
3. Quantos ciclos ocorreram em cada amostra?

## Montagem de genoma

1. Instalar (uma vez - https://github.com/ablab/spades):

```
mamba create -y -n montagem -c conda-forge -c bioconda spades=4.0.0 seqtk
```

2. Ativar o ambiente conda de montagem




```
nano scripts/02_montagem.sh
```

```
#!/bin/bash

mkdir -p resultados/02_spades
mkdir -p logs/02_spades

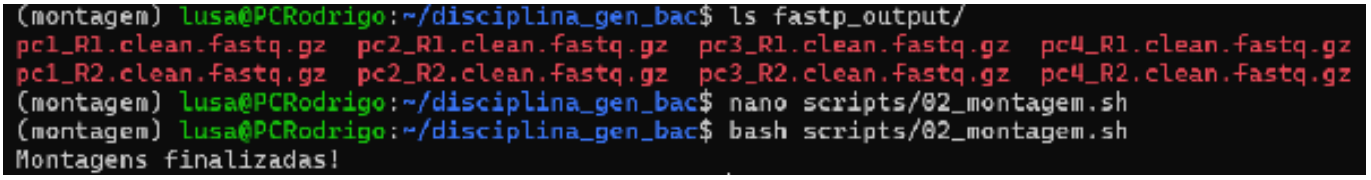
for sample in pc1 pc2 pc3 pc4
do
    spades.py \
```

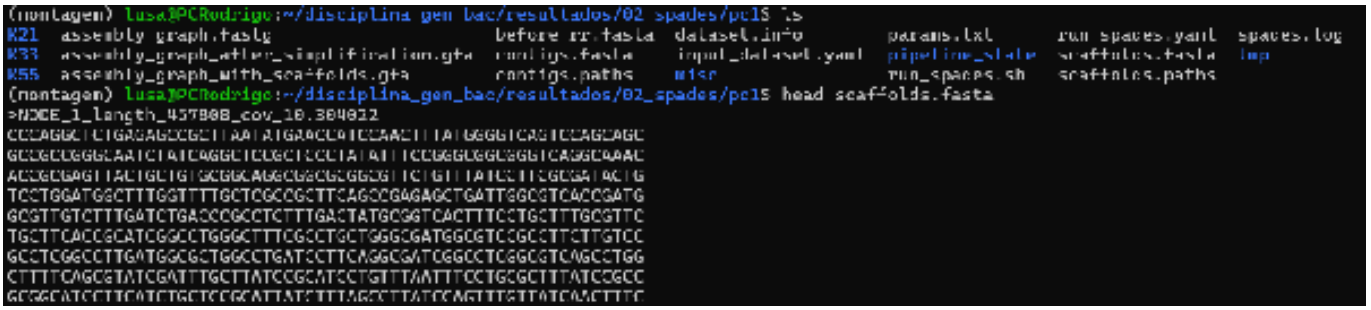
```
        --isolate \
        -1 fastp_output/${sample}_R1.clean.fastq.gz \
        -2 fastp_output/${sample}_R2.clean.fastq.gz \
        -o resultados/02_spades/${sample} \
        -t 4 \
        -m 8 \
        > logs/02_spades/${sample}.log 2>&1 &

done

wait

echo "Montagens finalizadas!"
```






3. Mover os scaffolds pra pasta dos genomas montados:

```
mkdir -p genomas_completos/spades

for sample in pc1 pc2 pc3 pc4
do
    mv resultados/02_spades/${sample}/scaffolds.fasta \
       genomas_completos/spades/${sample}_scaffolds.fasta
done
```

```
(montagem) lusa@PCRodrigo:~/disciplina_gen_bac$ mkdir -p genomas_completos/spades
(montagem) lusa@PCRodrigo:~/disciplina_gen_bac$ for sample in pc1 pc2 pc3 pc4
do
    mv resultados/02_spades/${sample}/scaffolds.fasta \
       genomas_completos/spades/${sample}_scaffolds.fasta
done
(montagem) lusa@PCRodrigo:~/disciplina_gen_bac$ ls genomas_completos/spades/
pc1_scaffolds.fasta  pc2_scaffolds.fasta  pc3_scaffolds.fasta  pc4_scaffolds.fasta
```

# Métricas de qualidade

## QUAST

1. Instalar (uma vez - https://github.com/ablab/quast):

```
mamba create -y -n quast -c conda-forge -c bioconda "python=3.11" quast
```

2. Ativar o ambiente e rodar:

```
conda activate quast
quast \
    genomas_completos/spades/pc1_scaffolds.fasta \
    genomas_completos/spades/pc2_scaffolds.fasta \
    genomas_completos/spades/pc3_scaffolds.fasta \
    genomas_completos/spades/pc4_scaffolds.fasta \
    -o resultados/03_quast \
    -t 4
```






Show heatmap

| Statistics without reference | pc1_scaffolds | pc2_scaffolds | pc3_scaffolds | pc4_scaffolds |
|---|---|---|---|---|
| # contigs | 149 | 149 | 34 | 6 |
| # contigs (>= 0 bp) | 471 | 471 | 69 | 52 |
| # contigs (>= 1000 bp) | 133 | 133 | 25 | 5 |
| # contigs (>= 5000 bp) | 84 | 84 | 20 | 4 |
| # contigs (>= 10000 bp) | 71 | 71 | 18 | 4 |
| # contigs (>= 25000 bp) | 50 | 50 | 18 | 4 |
| # contigs (>= 50000 bp) | 39 | 39 | 16 | 3 |
| Largest contig | 457 808 | 457 808 | 921 781 | 1 494 759 |
| Total length | 5 804 519 | 5 804 519 | 5 340 867 | 1 836 794 |
| Total length (>= 0 bp) | 5 851 446 | 5 851 446 | 5 349 708 | 1 845 341 |
| Total length (>= 1000 bp) | 5 792 558 | 5 792 558 | 5 334 651 | 1 835 813 |
| Total length (>= 5000 bp) | 5 649 897 | 5 649 897 | 5 317 511 | 1 831 156 |
| Total length (>= 10000 bp) | 5 548 689 | 5 548 689 | 5 300 211 | 1 831 156 |
| Total length (>= 25000 bp) | 5 221 528 | 5 221 528 | 5 300 211 | 1 831 156 |
| Total length (>= 50000 bp) | 4 824 392 | 4 824 392 | 5 239 047 | 1 791 253 |
| N50 | 116 506 | 116 506 | 399 553 | 1 491 759 |
| N90 | 24 801 | 24 801 | 124 756 | 148 089 |
| auN | 142 165 | 142 165 | 464 836 | 1 241 725 |
| L50 | 16 | 16 | 5 | 1 |
| L90 | 51 | 51 | 12 | 3 |
| GC (%) | 56.71 | 56.71 | 55 | 10.32 |
| **Per base quality** | | | | |
| # N's per 100 kbp | 8.96 | 8.96 | 7.68 | 27.22 |
| # N's | 520 | 520 | 410 | 500 |

## BUSCO

1. Instalar (uma vez - https://gitlab.com/ezlab/busco):

```
mamba create -y -n busco -c conda-forge -c bioconda busco
```

2. Criar a pasta do BUSCO

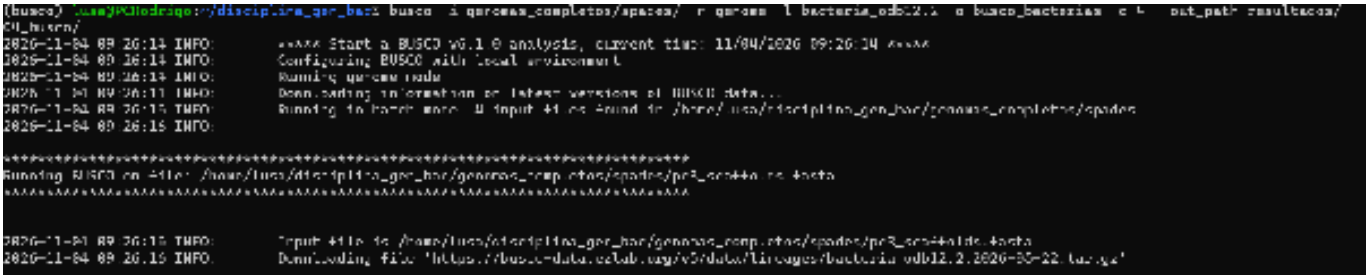
```
mkdir -p resultados/04_busco
```

3. Ativar o ambiente e rodar

```
conda activate busco

busco \
    -i genomas_completos/spades/ \
    -m genome \
    -l bacteria_odb12.2 \
    -o busco_bacterias \
```

```
    -c 4 \
    --out_path resultados/04_busco
```

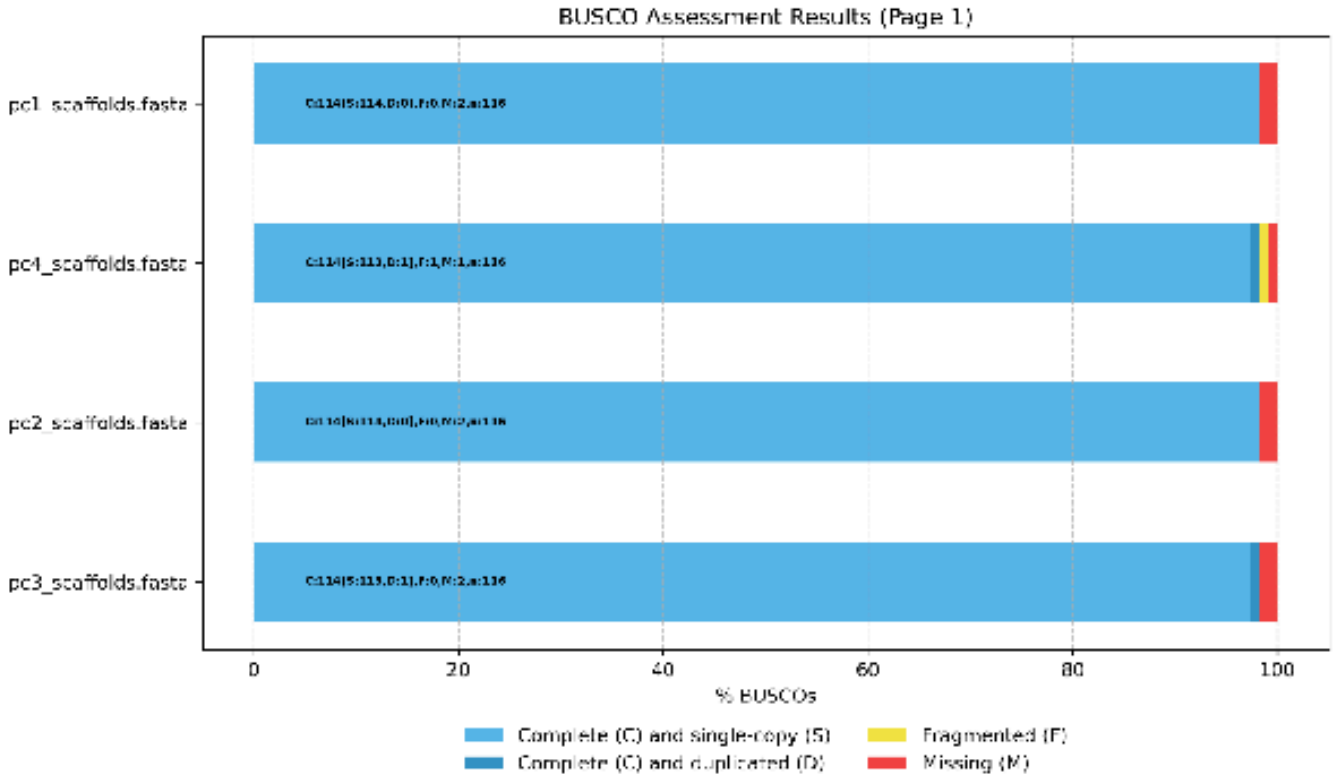


```
~/disciplina_gen_bac/resultados/04_busco/busco_bacterias$
```

```
batcat
pc1_scaffolds.fasta/short_summary.specific.bacteria_odb12.2.pc1_scaffolds.f
asta.txt
```

4. Plotagem

```
mkdir -p resultados/04_busco/plot
mv resultados/04_busco/busco_bacterias/* \
   resultados/04_busco/plot/
busco --plot resultados/04_busco/plot
```




# MOB-suite

1. Instalar (uma vez - https://github.com/phac-nml/mob-suite):

```
mamba create -y -n mob_suite -c conda-forge -c bioconda mob_suite
```

2. Ativar o ambiente e baixar o banco (se não baixou ainda):

```
conda activate mob_suite && mob_init     # baixa o banco, ~2 GB
```



3. Criar o script

```
nano scripts/
```

```
#!/bin/bash

mkdir -p resultados/05_mob logs/05_mob

for sample in pc1 pc2 pc3 pc4
do
    mob_recon \
        --infile genomas_completos/spades/${sample}_scaffolds.fasta \
        --outdir resultados/05_mob/${sample} \
        --num_threads 4 \
        --run_overhang \
        > logs/05_mob/${sample}.log 2>&1 &

done

wait

echo "MOB-recon finalizado!"
```

```
(mob_suite) lusa@PCRodrigo:~/disciplina_gen_bac$ nano scripts/05_mob.sh
(mob_suite) lusa@PCRodrigo:~/disciplina_gen_bac$ bash scripts/05_mob.sh
MOB-recon finalizado!
```

```
(mob_suite) lusa@PCRodrigo:~/disciplina_gen_bac/resultados/05_mob/pc1$ ls
biomarkers.blast.txt  plasmid_AA111.fasta  plasmid_AB714.fasta
chromosome.fasta      plasmid_AA338.fasta  plasmid_AC120.fasta
contig_report.txt     plasmid_AA405.fasta  plasmid_AD092.fasta
mge.report.txt        plasmid_AA452.fasta  plasmid_AE809.fasta
mobtyper_results.txt  plasmid_AA941.fasta
plasmid_AA048.fasta   plasmid_AA942.fasta
```

O que podemos ver?

1. Os elementos moveis (IS) - mge.report.txt

```
cut -f2,3,5,11,12,16,17,19,21 mge.report.txt | column -t -s$'\t'
```

| rotcclo_type | primary_cluster_id | contig_id | rga_type | mge_subtype | contig_start | contig_end | sstrand | pident |
|---|---|---|---|---|---|---|---|---|
| chromosome | | NODE_8_length_169619_cov_11.329268 | IS4pm28 | ISWE_ | 21868 | 22501 | | |
| chromosome | | NODE_58_length_15627_cov_12.069169 | TS1403 | TS3 | 10349 | 11553 | | |
| plasmid | AA405 | NODE_90_length_4113_cov_23.003705 | IS4pm26 | IS5 | 1024 | 2218 | minus | 98.745 |
| plasmid | AA405 | NODE_115_length_2279_cov_21.585103 | IS4pm21 | ISNL* | 1 | 2278 | minus | 99.649 |
| plasmid | AA048 | NODE_75_length_9177_cov_13.748679 | IS182 | IS6 | 3772 | 4825 | plus | 97.538 |
| plasmid | AA048 | NODE_75_length_9177_cov_13.748679 | IS3rssl | IS3 | 7876 | 9120 | minus | 98.876 |

2. Informações sobre os plasmídeos - mobtyper_results.txt

```
cut -f1,2,3,6,8,10,12,14 mobtyper_results.txt | column -t -s$'\t'
```

| sample_id | num_contigs | size | rep_type(s) | relaxase_type(s) | mpf_type | orit_type(s) | predicted_mobility |
|---|---|---|---|---|---|---|---|
| pc1_scaffolds:AA338 | 5 | 72815 | IncFIA,IncFIC | NOBF | MPF_F | NOBP | conjugative |
| pc1_scaffolds:AA435 | 29 | 231836 | IncFIB | - | MPF_F | NOBP | mobilizable |
| pc1_scaffolds:AA452 | 3 | 91391 | IncFIB,IncFII,Inc9 | NOBF,YOBF | MPF_F | - | conjugative |
| pc1_scaffolds:AE809 | 3 | 42427 | - | - | - | - | non-mobilizable |
| pc1_scaffolds:AB714 | 1 | 19171 | - | - | - | - | non-mobilizable |
| pc1_scaffolds:AA941 | 1 | 9349 | Col7NAI rep cluster 1057 | NOBC | | | mobilizable |
| pc1_scaffolds:AA048 | 2 | 12716 | IncFIA,rep_cluster_3927 | - | - | - | non-mobilizable |
| pc1_scaffolds:AC120 | 1 | 7848 | IncQ1 | NOBQ | - | NOBQ | mobilizable |
| pc1_scaffolds:AD092 | 2 | 5731 | - | - | - | - | non-mobilizable |
| pc1_scaffolds:AA942 | 1 | 1892 | - | - | - | - | non-mobilizable |
| pc1_scaffolds:AA111 | 1 | 1258 | - | - | - | - | non-mobilizable |

3. Hospedeiros observados de cada plasmídeo - mobtyper_results.txt

```
cut -f1,14,17,20,21 mobtyper_results.txt | column -t -s$'\t'
```

| sample_id | predicted_mobility | host_organism_name | predicted_host_range_overall_rank | predicted_host_range_overall_name |
|---|---|---|---|---|
| pc1_scaffolds:AA338 | conjugative | Citrobacter sp. CRE-16 | family | Enterobacteriaceae |
| pc1_scaffolds:AA405 | mobilizable | Escherichia coli | order | Enterobacterales |
| pc1_scaffolds:AA452 | conjugative | Escherichia coli | order | Enterobacterales |
| pc1_scaffolds:AE809 | non-mobilizable | Klebsiella pneumoniae | order | Enterobacterales |
| pc1_scaffolds:AB714 | non-mobilizable | Salmonella enterica subsp. enterica serovar Indiana | family | Enterobacteriaceae |
| pc1_scaffolds:AA941 | mobilizable | Klebsiella pneumoniae | order | Enterobacterales |
| pc1_scaffolds:AA048 | non-mobilizable | Klebsiella pneumoniae | order | Enterobacterales |
| pc1_scaffolds:AC120 | mobilizable | Klebsiella pneumoniae | phylum | Pseudomonadota |
| pc1_scaffolds:AD092 | non-mobilizable | Edwardsiella anguillarum ET080813 | order | Enterobacterales |
| pc1_scaffolds:AA942 | non-mobilizable | Klebsiella pneumoniae J96b | family | Enterobacteriaceae |
| pc1_scaffolds:AA111 | non-mobilizable | Klebsiella pneumoniae | genus | Klebsiella |

Movendo os novos arquivos fasta pra anotação:

```
mkdir -p genomas_completos/replicons

for pc in pc1 pc2 pc3 pc4; do
  for f in resultados/05_mob/$pc/chromosome.fasta
resultados/05_mob/$pc/plasmid_*.fasta; do
```

```
    mv "$f" "genomas_completos/replicons/${pc}__$(basename "$f")"
  done
done
```

## Como ficou o repositório no final?

```
.
├── Paired-end_data
│   ├── pc1
│   ├── pc2
│   ├── pc3
│   └── pc4
├── busco_downloads
│   └── lineages
├── fastp_input
├── fastp_output
├── genomas_completos
│   ├── replicons
│   └── spades
├── logs
│   ├── 02_spades
│   └── 05_mob
├── resultados
│   ├── 01_fastp
│   ├── 02_spades
│   ├── 03_quast
│   ├── 04_busco
│   └── 05_mob
│       ├── pc1
│       ├── pc2
│       ├── pc3
│       └── pc4
└── scripts
```